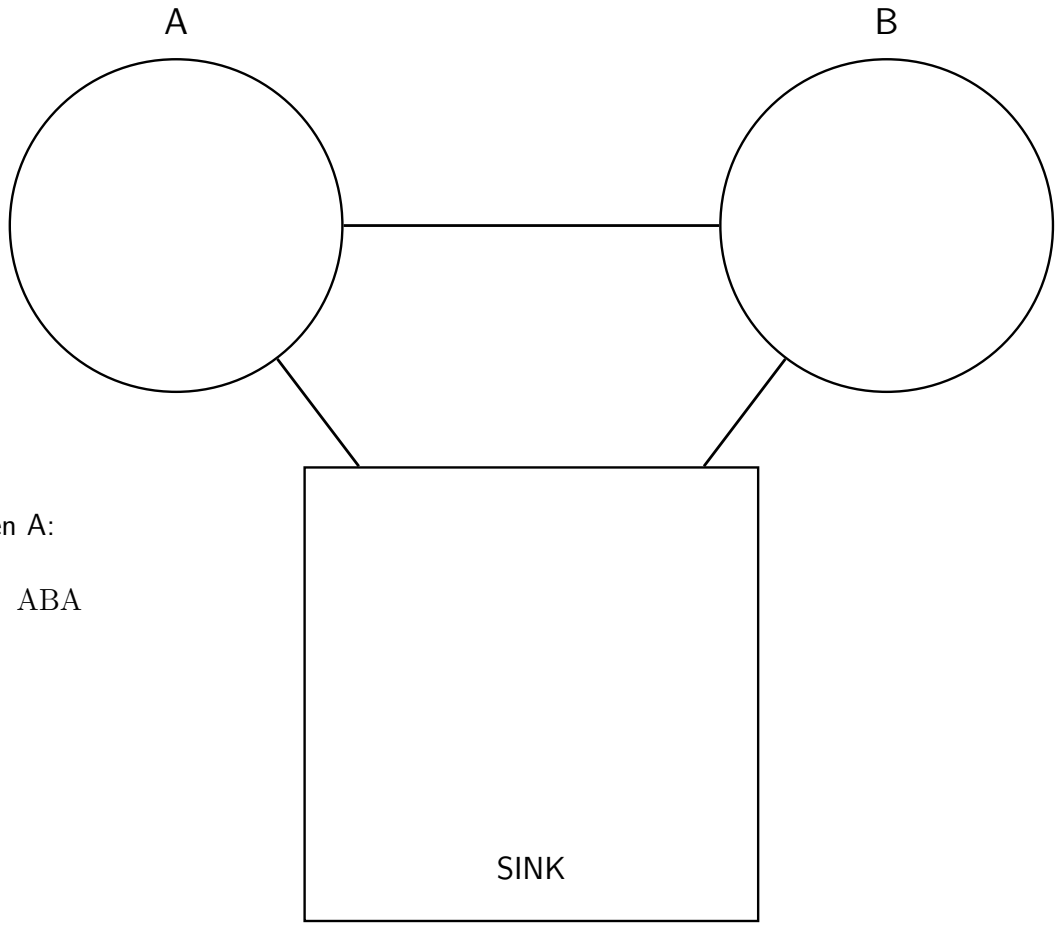
A circle needs one chip for every line touching it before it can share. To share, send one chip along each line. The sink keeps its chips and never shares. Choose one circle at a time. Stop when no circle can share. Empty the sink before each new start. Write each sharing letter in order, then the final piles.

Problem 1: Try each starting pair and share until you must stop. Can choosing a different order leave different piles?

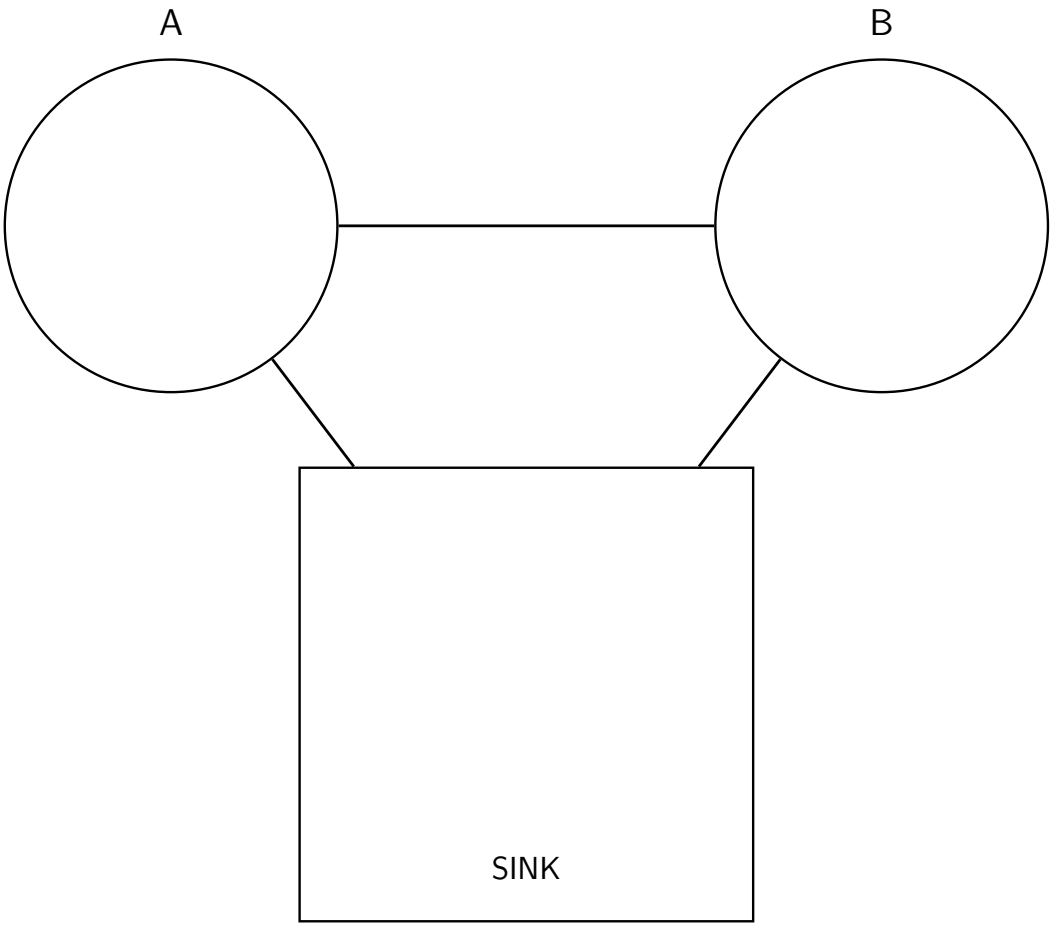


A, then B, then A:

$\mathrm{A} \rightarrow \mathrm{AB} \rightarrow \mathrm{ABA}$

Start A	Start B	Sharing word	Finish A	Finish B
● ●	● ●			
● ●	● ●			
● ● ●	● ●			
● ● ●	● ●			
● ●	● ● ●			
● ●	● ● ●			
● ● ●	● ● ●			
● ● ●	● ● ●			

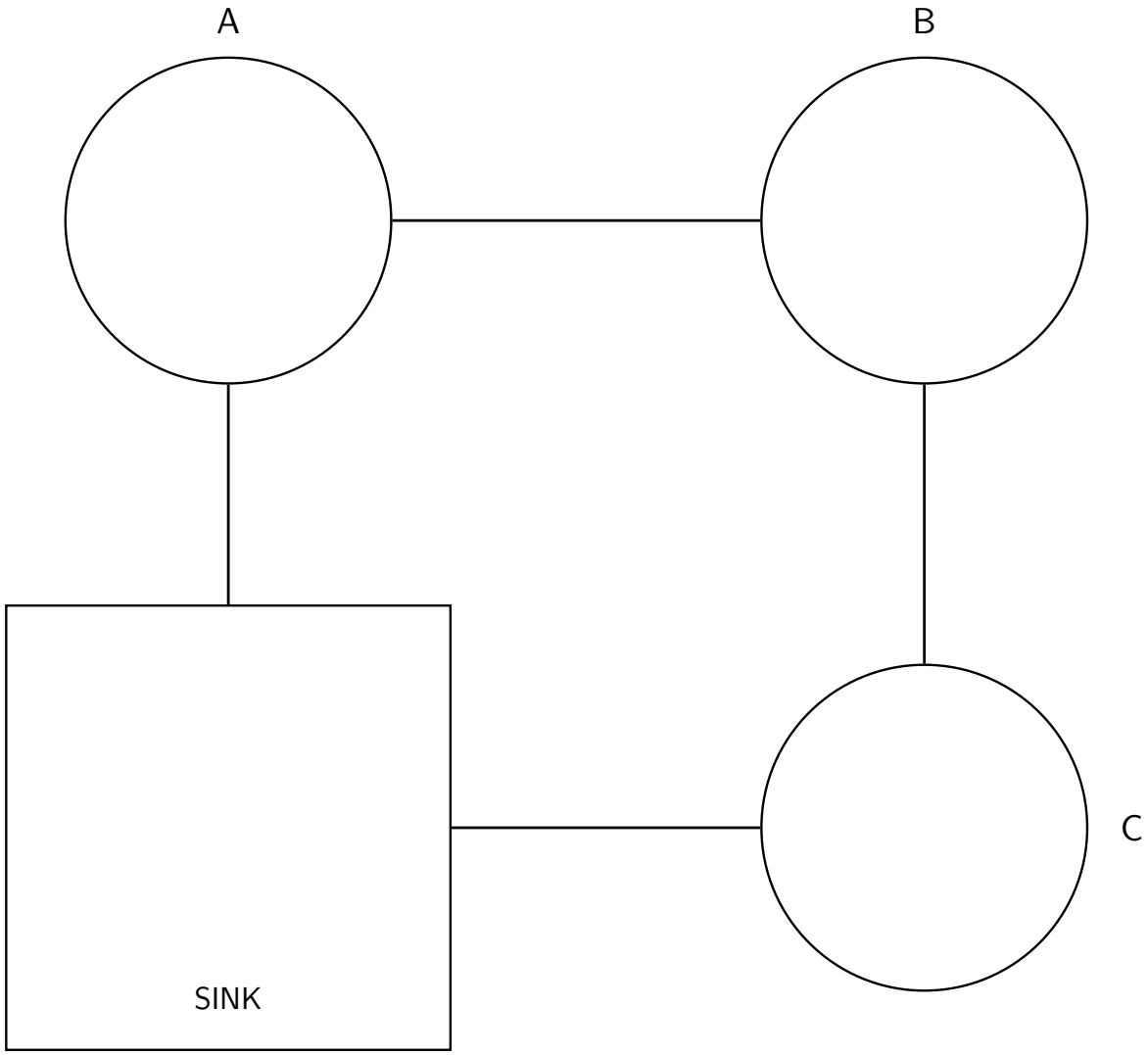
Problem 2: Put 4 chips altogether on A and B in any way, and share until you must stop. Find every finish; then try 5 chips.



4 chips

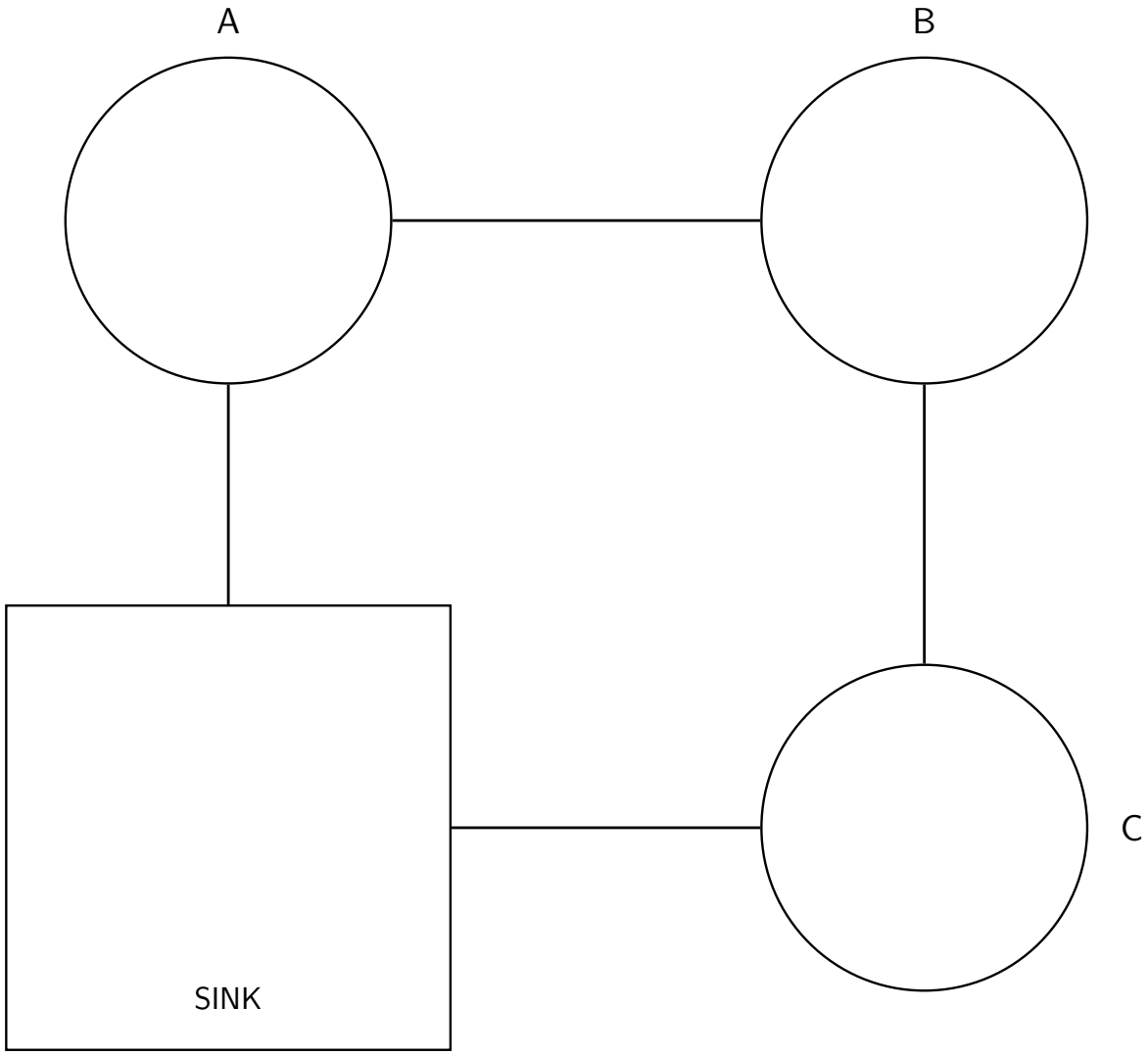
5 chips

Problem 3: Try each start and share until you must stop. Can choosing a different order leave different piles?



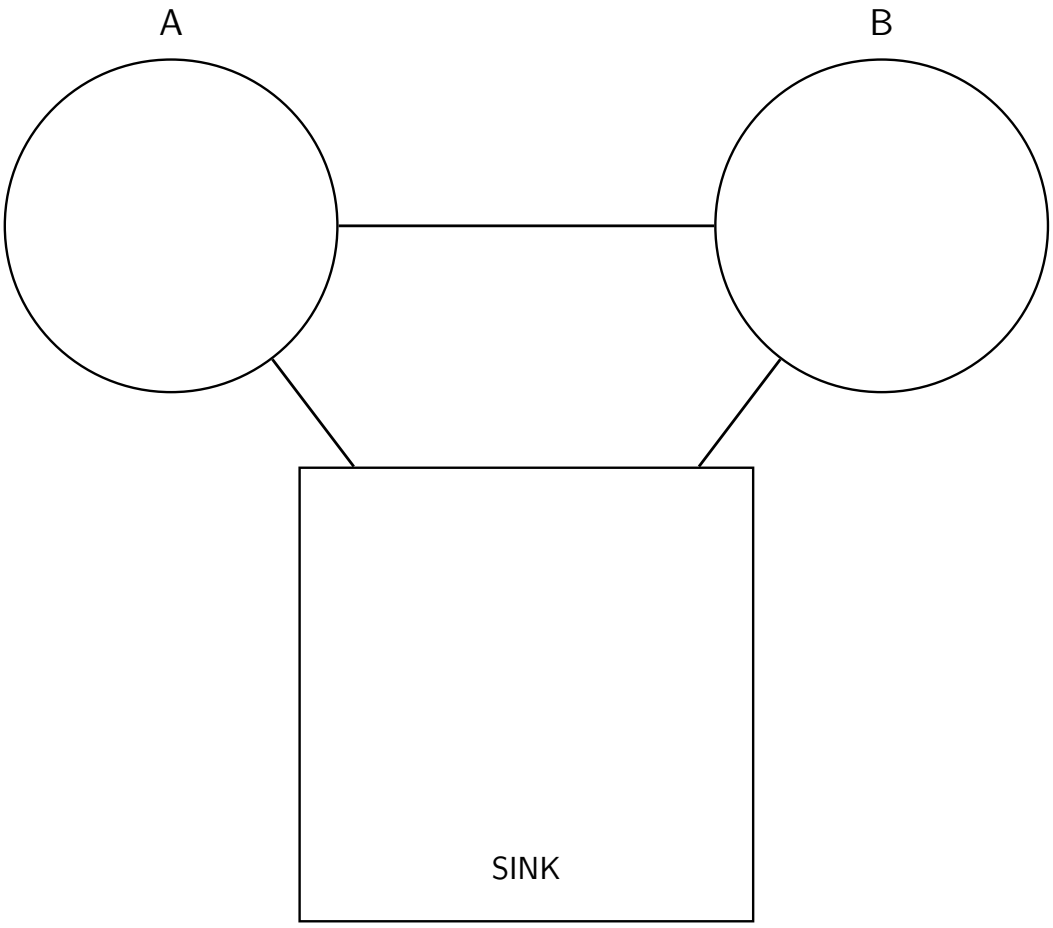
Start A	Start B	Start C	Sharing word	Finish A	Finish B	Finish C
0	● ● ● ●	0				
0	● ● ● ●	0				
● ●	● ●	0				
● ●	● ●	0				
0	● ●	● ●				
0	● ●	● ●				
● ●	0	● ●				
● ●	0	● ●				

Problem 4: Find every way to put chips on A, B, and C so that no circle can share. Can any of these ways use 4 chips?



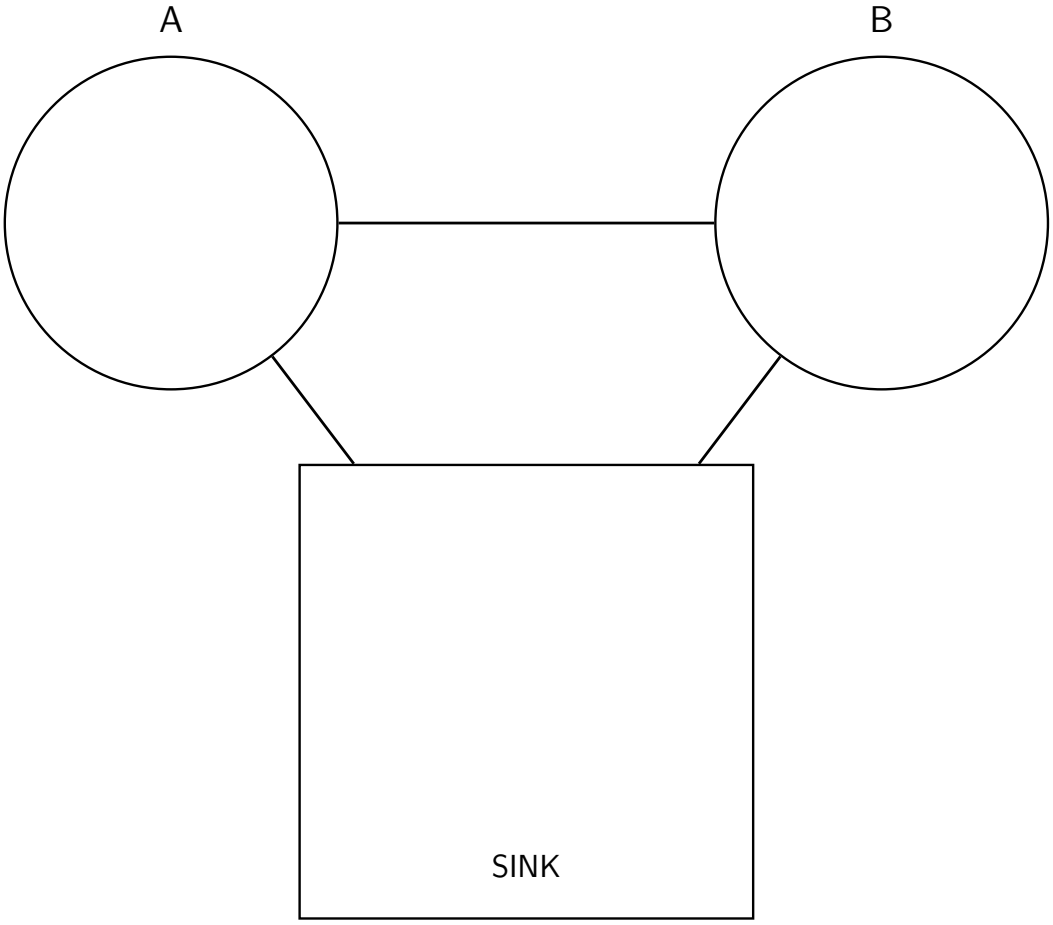
A	B	C

Problem 5: For each row, start empty and add the chips shown, sharing whenever you choose. Can a different order of adding and sharing change the finish?



Add at A	Add at B	Sharing word	Finish A	Finish B
● ● ●	● ● ●			
● ● ●	● ● ●			
● ● ●	● ● ●			
● ●	● ● ● ●			
● ●	● ● ● ●			
● ●	● ● ● ●			

Problem 6: Start with empty circles, then add one chip to A at a time, finishing all the sharing after each chip. Can both circles ever be empty again?



Chips added	Finish A	Finish B
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		